

David Xia

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EDUCATION

• University of Illinois Urbana-Champaign

August 2023 - May 2027

Bachelor of Science in Computer Science

Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration

Bachelor of Science in Liberal Arts & Sciences Major in Statistics

- GPA: 3.98/4.00

EXPERIENCE

• Illinois Combinatorics Lab for Undergraduate Experience (ICLUE) , Researcher

Jan 2024 - Present

- Mentored by Dr. Alexander Yong and Ph.D. student Duy Phan, studying algebraic combinatorics.
- Co-authored a paper on and currently researching the Robinson-Schensted-Knuth correspondence.

• National University of Singapore SEEDER Group , Researcher

May 2024 - July 2024

- Worked under guidance of Dr. Kelvin Xuanyao's SEEDER Group to developed GARDEN, a machine learning pipeline to detect, classify, and analyze plant diseases, improving previous accuracies by 40%.
- Implemented generative, classification, and segmentation models built utilizing PyTorch and TensorFlow framework, using self-collected data from local aquaculture farms, and presented findings in manuscript.

• University of California San Diego, Research Intern

June 2020 - August 2023

- Worked with Dr. Xinyu Zhang and his PhD students on various projects over 4 years.
- Collaborated with Dr. Ke Sun to develop EveGuard, a software framework that defends against smartphone side-channel eavesdropping using an adversarial perturbation generator model while maintaining audio quality.
- Trained segmentation and classification models for CogTeach, a facial expression recognition application tool for virtual learning, and conducted pilot study by delivering courses of lectures on neural networks.

PROJECTS

• Predictability and Competitiveness of Sports Leagues, Python

January 2024 - Present

- Researched the data science perspective of sports with Dr. AJ Hildebrand at the Illinois Mathematics Lab.
- Built web scrapers with Selenium to scrape online data and analyzed the predictability of sports matches by comparing prediction accuracies of multiple sources including polls and the betting market.
- Evaluated efficiency of model based probabilistic predictions.
- Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, Joint Mathematics Meeting 2025.

• GARDEN: ML pipeline for anomaly detection, PyTorch, TensorFlow

May 2024 - July 2024

- Machine learning pipeline project at the NUS SEEDER Group.
- Developed machine learning pipeline to detect, classify, and analyze diseases found in plant leaves.
- Trained U-Net segmentation models and classification models based on convolutional neural networks to isolate and classify images using the TensorFlow framework, achieving 90% segmented classification rate.
- Generated synthetic images using custom trained cGAN model to increase data variety and volume.
- Stacked deep learning models with traditional techniques like support vector machines, gradient boosting, logistic regression, and decision trees to achieve an increased detection accuracy of over 95%.

• EveGuard: Software framework for eavesdropping defense, TensorFlow, GAN

May 2022 - August 2023

- Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.
- Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
- Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
- Co-authored a paper due to appear in the IEEE Symposium on Security and Privacy 2025.

PRESENTATIONS

- 2025 **Algecom** RSK linear operators and the Vershik-Kerov-Logan-Shepp curve (presented by Duy Phan)
- 2025 **Joint Mathematics Meeting Pi Mu Epsilon Poster Session** Predictability in Major Sports Leagues
- 2024 **Illinois Mathematics Lab Research Symposium** Predictability in College Sports
- 2024 **University of Illinois Undergraduate Research Symposium** Predictability in College Sports
- 2024 **Rose-Hulman Undergraduate Math Conference** Predictability in College Sports

PAPERS

PRE=PREPRINT, ACC=ACCEPTED

- [1] **[PRE] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve** [[arXiv](#)]
Duy Phan, David Xia
Preprint
- [2] **[ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations** [[arXiv](#)]
Jung-Woo Chang, Ke Sun, David Xia, Xinyu Zhang, Farinaz Koushanfar
To appear in IEEE Symposium on Security and Privacy, 2025